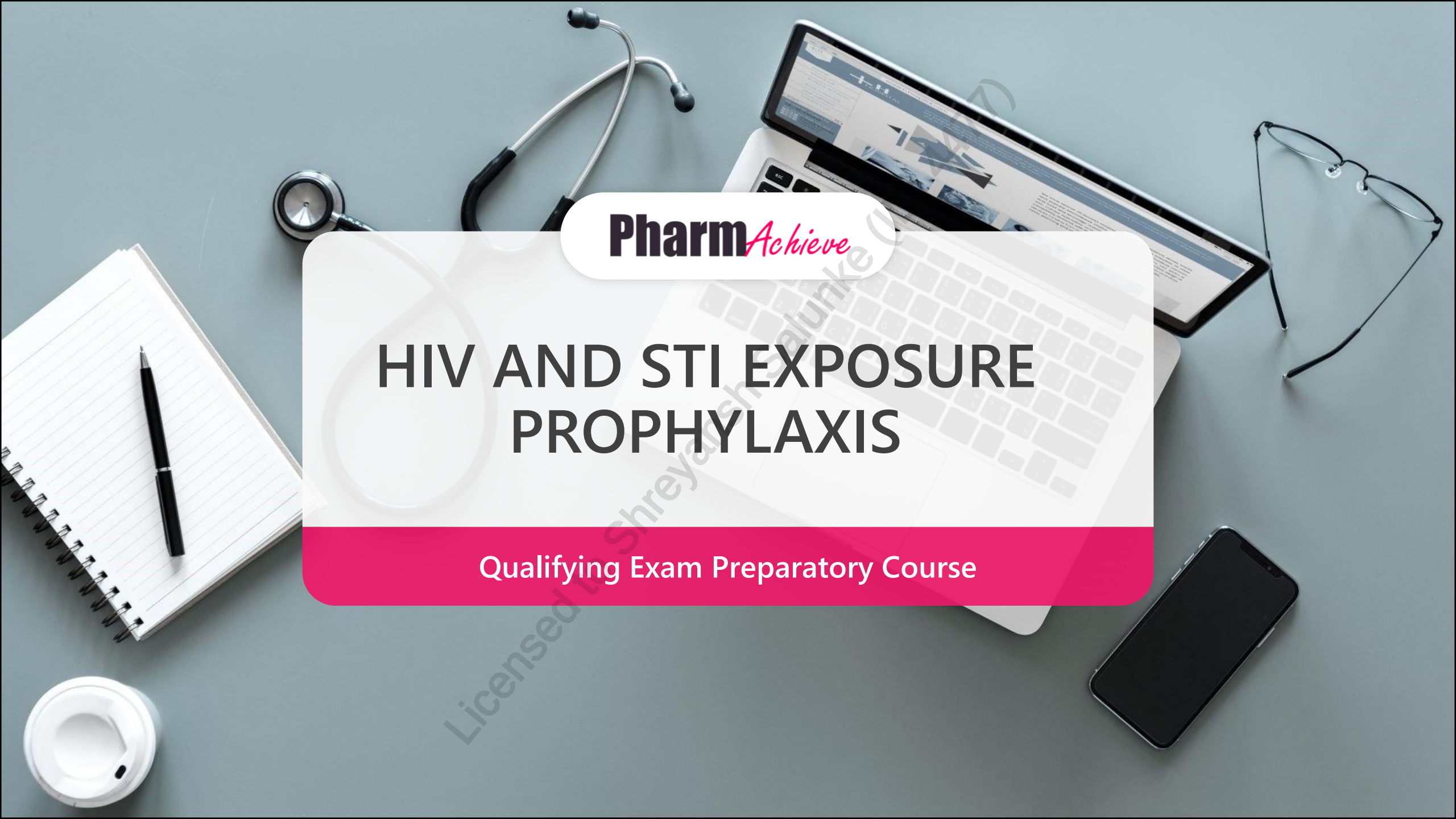


A top-down view of a grey desk with various items: a stethoscope, a laptop with a tablet on top, a pair of glasses, a smartphone, a spiral notebook with a pen, and a coffee cup. A large white pill is faintly visible in the background.

Pharm*Achieve*

HIV AND STI EXPOSURE PROPHYLAXIS

Qualifying Exam Preparatory Course

COMPETENCIES COVERED IN THIS PRESENTATION...

**Providing Care:
Clinical Care**

**Providing Care:
Distribution**

**Knowledge
& Expertise**

**Communication
& Collaboration**

**Leadership &
Stewardship**

Professionalism

LEGEND

BMD Bone Mineral Density

CBC Complete Blood Count

GBMSM Gay, Bisexual, and Men who have Sex with Men

HBsAg Hepatitis B Surface Antigen

nPEP Non-occupational Post-Exposure Prophylaxis

oPEP Occupational Post-Exposure Prophylaxis

PrEP Pre-exposure Prophylaxis

PWID Persons Who Inject Drugs

SCr Serum Creatinine

DEFINITIONS

Pre-exposure Prophylaxis (PrEP)

Daily or episodic antiretroviral therapy in those at high risk of HIV acquisition

Non-occupational Post-Exposure Prophylaxis (nPEP)

The administration of antiretroviral therapy **after a potential transmission event** has occurred

Occupational Post-Exposure Prophylaxis (PEP)

The administration of antiretroviral therapy **after a potential transmission event has occurred, usually within a health care environment** and involving percutaneous exposure

RISK FACTORS

Risk of HIV Transmission Determined by Viral Load, Other Disease, and Risk Group

- **Undetectable = Untransmissible (U=U)**
- Those with HIV detectable viral loads, risk of transmission determined by concurrent sexually transmitted infection (STI) as well as additional barrier methods used
- Other STIs are an independent risk factor for viral transmission

Substantial

- HIV+ and viremic (i.e. viral load >40 copies/mL)
- HIV status unknown, source from population with high prevalence

Low but Nonzero

- HIV+ with undetectable viral load but with concurrent STI

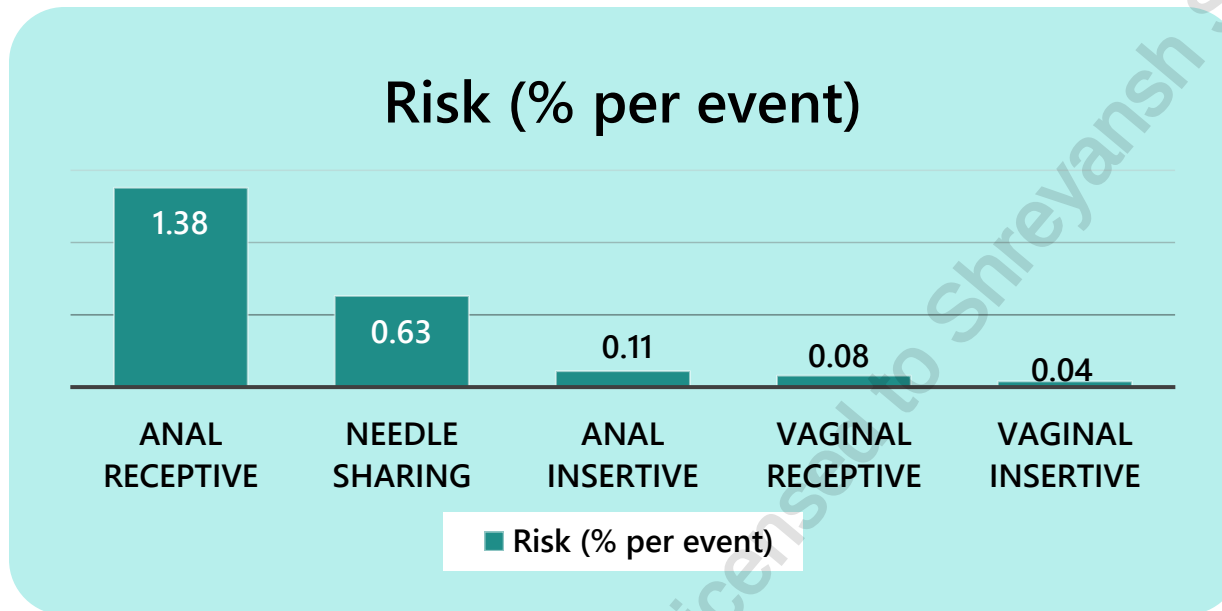
Negligible or None

- HIV+, undetectable viral load and no STI
- General population

RISK FACTORS

Risk of HIV Transmission by Behaviour or Action

- **High risk: anal receptive sex, needle sharing**
- Moderate risk: anal insertive, vaginal sex (insertive/receptive)
- Low risk: oral sex, sex toys, blood on compromised skin



PRE-EXPOSURE PROPHYLAXIS

Pre-exposure Prophylaxis (HIV PrEP) Eligibility

- **GBMSM or transgender women engaging in condomless anal sex in the last 6-month period and (if any of):**
 - Syphilis, rectal STI in last 12 months
 - Recurrent nPEP use
 - Ongoing relationship with HIV+ partner with risk of transmission
 - High-incidence risk index ≥ 11
- **Heterosexual (if any of):**
 - Ongoing relationship with HIV+ partner with substantial risk of transmission
 - Consider in those with relationships with HIV+ with low but non-negligible risk
- **Persons Who Inject Drugs (PWID)**
 - Consider in those who share needles with HIV+ person with non-negligible risk

PRE-EXPOSURE PROPHYLAXIS

Pre-exposure Prophylaxis (HIV PrEP) Regimens

(note: evidence/drug preference varies by population)

- Primary regimen (one of):
 - Tenofovir DF 300 mg/Emtricitabine 200 mg PO daily (all patients)
 - or
 - Tenofovir AF 25 mg/Emtricitabine 200 mg PO daily (cis men, transwomen engaged in sex with cis man, consider in others)
 - or
 - Cabotegravir 600mg IM (qmonth>q2monhts) (cis men, trans women engaged in sex with cis man, consider in others)
- Alternative regimen
 - “PRN regimen” Tenofovir DF 300mg/Emtricitabine 200 mg, 2 tabs PO once within 2-24 hours prior to exposure followed by 1 tablet daily until 48 hours after the last exposure (cis men, trans women engaged in sex with cist man, not recommended in cis gendered women)

ASSESSMENT AND MONITORING

Baseline assessment (for HIV PrEP)

- HIV screening
- STI screening
- SCr/Urinalysis
- BMD assessment is not necessary but should be considered in those at risk of OP
- bHCG
- Hepatitis A, B, C screening
- CBC

Follow up monitoring (for PrEP)

- **First 30 days:** HIV, SCr, adherence, and counseling
- **Every 3 months:** SCr, bHCG, HIV, STIs, adherence, and counseling
- **Every 12 months:** Hepatitis B, C screening

POST-EXPOSURE PROPHYLAXIS

Non-Occupational HIV Post-Exposure Prophylaxis (nPEP) Eligibility:

- Recommended if HIV negative person presenting within 72 hours of:
 - Moderate or high-risk exposure with a person with substantial risk for HIV transmission
- Consider if HIV negative person presenting within 72 hours of:
 - Moderate or high-risk exposure with a person with low but non-negligible risk for HIV transmission

POST-EXPOSURE PROPHYLAXIS

Non-Occupational HIV Post-Exposure Prophylaxis (nPEP) Preferred Regimens (choose one):

- Tenofovir AF 25 mg/Emtricitabine 200 mg/Biktegravir 50 mg PO daily x 28 days
- Tenofovir DF 300 mg/Emtricitabine 200 mg PO daily + Dolutegravir 50 mg PO daily x 28 days

Consider “pill-in-pocket” for patients at episodic high risk encounters

Start as soon as feasible and dispense the entire course (if possible) on initial dispensing event

POST-EXPOSURE PROPHYLAXIS

Non-Occupational Post-Exposure Prophylaxis (nPEP) Alternative Regimens:

Avoid use of the following regimens:

- Elvitegravir based regimens
- Raltegravir/TAF regimens
- Rilpivirine based regimens

ASSESSMENT AND MONITORING

Baseline assessment (for HIV nPEP)

- HIV screening
- STI screening
- SCr/Urinalysis
- bHCG
- Hepatitis A,B,C screening
- CBC
- ALT

Follow up monitoring (for nPEP)

- Week 2: SCr, ALT (only if abnormal at baseline/symptomatic)
- Week 12: HIV, Hepatitis C, STI

HIV PREVENTION AND RISK REDUCTION STRATEGIES

Don't delay

- Earlier PEP (immediate to < 72 hours) and well considered PrEP=better outcomes

Never waste an opportunity

- Barrier methods still work, counsel on them
- Safe sex practices are still important to reduce risk
- Needle exchanges work if provided without barriers
- Follow-up for other STI and health issues when you have a patient in front of you

Antiretrovirals are expensive

- Ensure coverage
- Evaluate options for lower cost therapy
- Investigate compassionate use programs

POST-EXPOSURE PROPHYLAXIS

Bacterial STI PEP (DoxyPEP)

- Recommended for GBMSM presenting within 72 hours of unprotected intercourse
- Reduces risk of syphilis, gonorrhea and chlamydia by 60%
 - 81% for chlamydia
 - 77% for syphilis
 - 41% for gonorrhea
- Dose: doxycycline 200mg PO x 1 dose
- Counselling:
 - Drug interactions
 - Incomplete efficacy
 - Need for comprehensive STI screening

POST-EXPOSURE PROPHYLAXIS:

FYI

Occupational Post-Exposure Prophylaxis (PEP) Eligibility:

High Risk

- Percutaneous exposure to potentially infected substances (blood, CSF, vaginal secretions, semen, synovial, peritoneal, pleural, ascitic, amniotic fluid)

Moderate Risk

- Mucous membrane exposure to potentially infectious substances
- Exposure of broken skin to potentially infectious fluids
- Bite wounds

Low risk

- Intact skin
- Exposure to low-risk fluids (urine, feces etc.)

POST-EXPOSURE PROPHYLAXIS:

FYI

Occupational Post-Exposure Prophylaxis (PEP) Eligibility:

If moderate to high-risk exposure to potentially infectious substances, **consider source HIV risk**

- Source of exposure
 - HIV + or HIV status unknown but epidemiology suggestive: Treat and test source
 - HIV unknown, source unavailable for testing/assessment: case-by-case

POST-EXPOSURE PROPHYLAXIS

Occupational HIV Post-Exposure Prophylaxis (nPEP) Preferred Regimens (choose one):

- Tenofovir AF 25 mg/Emtricitabine 200 mg/Biktegravir 25mg PO daily x 28 days
- Tenofovir DF 300 mg/Emtricitabine 200 mg PO daily + Dolutegravir 50 mg PO daily x 28 days

Start as soon as feasible and dispense the entire course (if possible) on initial dispensing event

POST-EXPOSURE PROPHYLAXIS

Occupational Post-Exposure Prophylaxis (nPEP) Alternative Regimens:

Avoid use of the following regimens:

- Elvitegravir based regimens
- Raltegravir/TAF regimens
- Rilpivirine based regimens

POST-EXPOSURE PROPHYLAXIS:

FYI

Occupational Post-Exposure Prophylaxis (PEP) Alternative Regimens:

- Zidovudine 300 mg/Lamivudine 150 mg PO BID x 28 das
- OR
- Tenofovir DF 300 mg + Lamivudine 300 mg PO daily x 28 days

PLUS (one of):

- Atazanavir 300 mg PO daily + Ritonavir 100 mg PO daily x 28 days
- Lopinavir 800 mg/Ritonavir 200 mg PO daily x 28 days
- Raltegravir 1200 mg PO daily x 28 days
- Etravirine 200 mg PO BID x 28 days
- Rilpivirine 25 mg PO daily x 28 days

ASSESSMENT AND MONITORING

FYI

Baseline assessment (for PEP)

- HIV screening
- SCr/Urinalysis
- bHCG
- Hepatitis A,B,C screening
- CBC
- ALT

Follow up monitoring (for PEP)

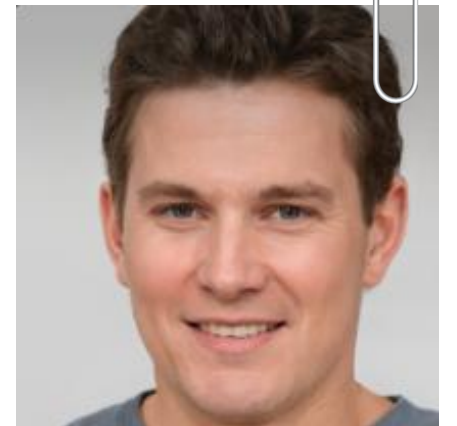
- Week 2: SCr, ALT (only if abnormal at baseline/symptomatic)
- Week 12: HIV, Hepatitis C

TAKEAWAY

- **Risk factors** for HIV transmission include anal receptive/insertive sex, needle sharing, blood on compromised skin, oral sex, vaginal insertive/receptive sex, and shared sex toys
- Eligibility for prophylaxis is determined by the **source of transmission** and the level of **risk of transmission** (low, moderate, high)
- Baseline assessment includes screening for **HIV, STIs, Hepatitis A, B, and C** along with laboratory testing for **CBC, ALT, SCr, bHCG**
- Pre-exposure prophylaxis regimens include: Tenofovir AF/DF + emtricitabine or cabotegravir
- Post-exposure prophylaxis regimens (both occupational and non-occupational) include:
 - Tenofovir DF or AF + Emtricitabine + INSTI or boosted PI

CASE 1

MS is a 24-year-old male who presents to the emergency department within 30 hours of a sexual encounter. He reports condomless receptive anal intercourse with a new male partner of unknown HIV status. He reports no symptoms and has no known medical history. He is not taking any regular medications, including no current pre-exposure prophylaxis (PrEP) regimen. MS is currently completing his fourth and final year of the psychology program at university and is living in residency with a classmate. He aspires to become a psychotherapist upon graduating. MS is not taking any medications and denies current recreational drug use or intravenous drug use. He smokes socially (1-2 cigarettes per day, 2-3 times a week) and drinks alcohol occasionally (2-3 standard drinks on weekends). He indicates he has no known drug allergies. On assessment, his rapid HIV test is negative. Baseline laboratory results show: creatinine clearance (CrCl) 105 mL/min, ALT 21 U/L, HBsAg negative, anti-HBs positive, anti-HCV negative. He expresses concern about HIV exposure and asks about post-exposure prophylaxis (PEP), as well as how to prevent future risks.



CASE 1 - QUESTIONS

1. What is the most appropriate next step regarding HIV post-exposure prophylaxis (PEP) in this patient?
 - a) No PEP needed; initiate PrEP today
 - b) Begin 3-drug PEP regimen immediately and reassess HIV status at 12 weeks
 - c) Begin 2-drug PEP regimen and reassess in 3 days
 - d) Delay PEP; repeat HIV test in 7 days before deciding
2. Which of the following is the recommended PEP regimen according to CMAJ 2025 guidance?
 - a) Emtricitabine/tenofovir alafenamide + rilpivirine
 - b) Zidovudine/lamivudine + efavirenz
 - c) Emtricitabine/tenofovir disoproxil fumarate + dolutegravir
 - d) Abacavir/lamivudine + nevirapine
3. When can this patient be considered for initiation of PrEP, assuming follow-up HIV testing remains negative?
 - a) Immediately after completing PEP
 - b) 4 weeks after completing PEP
 - c) Only if another exposure occurs
 - d) After a 3-month waiting period
4. Which lab parameter supports the safe initiation of tenofovir-based PEP and eventual PrEP use in this patient?
 - a) Negative HBsAg
 - b) Anti-HBs positivity
 - c) ALT of 21 U/L
 - d) Creatinine clearance of 105 mL/min

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CHANGE LOG

July 2025

- Amended title to include nPEP and PrEP (STI+HIV)
- Slide 3: Added legend
- Amended definitions to include GBMSM to be consistent with terminology
- Slide 8: Added cabotegravir
- Added slide 15 on bacterial STI PEP
- Added case practice
- Marked oPEP slides as FYI

October 2025

- No content changes made

March 2026

- Major updates following 2025 Canadian PEP/PrEP guidelines